

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs only on local server	Limited accessibility	High
2	No user authentication	Limited security	Medium
3	Uses static dataset	Needs periodic updates	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local usage	Deploy on cloud platform
2	Data Storage	CSV dataset	MySQL or MongoDB integration
3	Performance	Single server	Cloud hosting with better scalability
4	Security	Basic Flask application	HTTPS and user authentication

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment (Render)	1 Month	Public accessibility

Phase 2	Weather API Integration	2 Months	More accurate recommendations
Phase 3	Android Mobile Application	6 Months	Improved accessibility